

Submission Worksheet

Submission Data

Course: IT202-008-S2026

Assignment: IT202 PHP Challenge Scenarios

Student: Ashton T. (ast56)

Status: Submitted | **Worksheet Progress:** 100%

Potential Grade: 10.00/10.00 (100.00%)

Received Grade: 0.00/10.00 (0.00%)

Started: 2/24/2026 7:04:43 PM

Updated: 2/24/2026 8:27:50 PM

Grading Link: <https://learn.ethereallab.app/assignment/v3/IT202-008-S2026/it202-php-challenge-scenarios/grading/ast56>

View Link: <https://learn.ethereallab.app/assignment/v3/IT202-008-S2026/it202-php-challenge-scenarios/view/ast56>

Instructions

- Overview Link: <https://youtu.be/iwA6vyRPhoM> (older reference, but should be accurate for now)
- 1. Ensure you read all instructions and objectives before starting. It's recommended to solve all the requirements first, then fill out the worksheet.
- 2. Create a new branch from qa called M2-Homework
 - 1. `git checkout qa` (ensure proper starting branch)
 - 2. `git pull origin qa` (ensure history is up to date)
 - 3. `git checkout -b M2-Homework` (create and switch to branch)
- 3. Copy the template code from here: [GitHub Repository - M2 Homework](#)
 - It includes Scenarios 1-4 and `base.php`. Put all into an M2 folder or similar inside your `public_html`
 - Immediately record to history
 - `git add public_html`
 - `git commit -m "adding M2 HW baseline files"`
 - `git push origin M2-Homework`
 - Create a Pull Request from M2-Homework to qa and keep it open
- 4. Fill out the below worksheet
 - Each Scenario requires the following as you work
 - Ensure there's a comment with your UCID, date, and brief summary of how the problem was solved
 - Initial outline/plan of how you'll solve it via comments (add/commit after this stage)
 - Code solution (add/commit periodically as needed)
 - If you chose to tackle this problem by problem and fill out the worksheet as you go, you'll likely need to keep merging pull requests into qa
- 5. Once finished, click "Submit and Export"
- 6. Locally add the generated PDF to a folder of your choosing inside your repository folder (homework folder is fine) and move it to Github
 - 1. `git add .`
 - 2. `git commit -m "adding PDF"`

3. git push origin M2-Homework
4. On Github merge the pull request from M2-Homework to qa
5. On Github create a pull request from qa to prod and immediately merge. (This will trigger the prod deploy to make the Render.com prod links work)
7. Upload the same PDF to Canvas
8. Sync Local
 1. git checkout qa
 2. git pull origin qa

Section #1: (2 pts.) Scenario 1 - Odds

Progress: 100%

≡ Task #1 (2 pts.) - Edit the `printOdds` function to output odd values of the array

Progress: 100%

Details:

- Only make edits where noted via provided comments
- Challenge 1: From each passed in array, print odd values only in a single line separated by commas and a space after each comma (should not have leading or trailing commas)
- Step 1: sketch out plan using comments (include ucid and date)
- Step 2: Add/commit your outline of comments (required for full credit)
- Step 3: Add code to solve the problem (add/commit as needed)

📁 Part 1:

Progress: 100%

Details:

Two screenshots are expected

1. Snippet of relevant code showing solution (with ucid/date comment present)
2. Full output of executing the program (visit the proper file on Render.com qa after a manual deploy)

```
$output_result = "";
// Start Solution Edits
// set solution to $output result variable
//Avtbb 02/23/25

//Step 1: Use foreach command and a loop to iterate through array and divide values by 2
foreach ($arr as $odd){
    if ($odd % 2 != 0){
        $output_result .= $odd . ", ";
    }
}

//Step 2: Use rtrim to remove leading and trailing commas
$output_result = rtrim($output_result, ", ");
| You, yesterday - Added code to steps
// End Solution Edits
printScenarioOutput($output_result);
echo "</div>";
```

Code snippet problem 1

Objective: Find out odd and even values in a single line executed by command

Array: 1

Original Array:

0, 1, 2, 3, 4, 5, 6, 7, 8, 9

Output Array:

1, 3, 5, 7, 9

Array: 2

Original Array:

0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19

Output Array:

1, 3, 5, 7, 9, 11, 13, 15, 17, 19

Array: 3

Original Array:

0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19

Output Array:

0, 2, 4, 6, 8, 10, 12, 14, 16, 18

Array: 4

Original Array:

0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99

Output Array:

0, 2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30, 32, 34, 36, 38, 40, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, 64, 66, 68, 70, 72, 74, 76, 78, 80, 82, 84, 86, 88, 90, 92, 94, 96, 98, 100

Completed Scenario 1 for [ast56] [2026-02-25 00:01:43]

Render Output Problem 1

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Part 2:

Progress: 100%

Details:

- Direct link to the file in the homework related branch from Github (should end in `.php` , , zero credit if it does not)
- Direct link to the file on Render.com Prod (Grab the qa url you're testing, paste it, change "-qa" to "-prod"), zero credit if

URL #1

https://github.com/PanicGeneral/ast56-it202-008-qa/public_html/M2/scenario1.php



URL

<https://github.com/PanicGeneral/>



URL #2

<https://ast56-it202-008.onrender.com/M2/scenario1.php>



URL

<https://ast56-it202-008.onrender.com/M2/scenario1.php>



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Part 3:

Progress: 100%

Details:

Briefly explain how your solution works. Describe the logic, steps, or decisions your code makes to reach the result. Do not restate the problem or list requirements.

Your Response:

I solved this by first using the foreach command to iterate through each item in the array. Those values were then placed into the variable \$oddd and divided by 2. If the result of that division was not 0 then it was determined that it was odd and then saved to the variable \$output_result. I then used rtrim on the \$output_result to remove the leading and trailing commas.

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Section #2: (2 pts.) Scenario 2 - Sum

Progress: 100%

≡ Task #1 (2 pts.) - Edit the `sumValues` function to sum the array values and present them in a format with exactly two decimal places

Progress: 100%

Details:

- Only make edits where noted via provided comments
- Challenge 1: Sum all the values of the passed in array and assign to the `total` variable
- Challenge 2: Have the sum (total) be represented as a number with exactly 2 decimal places (similar to currency), assign to `modifiedTotal` variable
- Example: 0.1 would be shown as 0.10, 1 would be shown as 1.00, 0.011 as 0.01, etc
- Step 1: sketch out plan using comments (include ucid and date)
- Step 2: Add/commit your outline of comments (required for full credit)
- Step 3: Add code to solve the problem (add/commit as needed)

Part 1:

Progress: 100%

Details:

Two screenshots are expected

1. Snippet of relevant code showing solution (with ucid/date comment)
2. Full output of executing the program (visit the proper file on Render.com qa after a manual deploy)

```
$total = 0;
// Start Solution Edits
//ast56 02/23/26
// Solve Challenge 1 here: Sum all values
//Step 1: Iterate through array and add values and assign that sum to the variable total
$total= array_sum($arr);

// Solve Challenge 2 here: Format to 2 decimal places
//Step 3: Use the number_format to round the value of the totals to 2 decimal places.
$modifiedTotal = "?";
$modifiedTotal = number_format($total,2,'.','');
// End Solution Edits
printScenario2Output($total, $modifiedTotal);
echo "</div>";
```

Code snippet of Problem 2

Completed Scenario 2 For [ast56] [2026-02-23 00:02:02]

Output: [ast56] [2026-02-23 00:02:02]

```
Scenario 2: Sum
Array (
    0 => 1
    1 => 2
    2 => 3
    3 => 4
    4 => 5
    5 => 6
    6 => 7
    7 => 8
    8 => 9
    9 => 10
)
Sum: 55
Modified Total: 55.00
```

Scenario 2: Sum
Array (
 0 => 1
 1 => 2
 2 => 3
 3 => 4
 4 => 5
 5 => 6
 6 => 7
 7 => 8
 8 => 9
 9 => 10
)
Sum: 55
Modified Total: 55.00

Scenario 2: Sum
Array (
 0 => 1
 1 => 2
 2 => 3
 3 => 4
 4 => 5
 5 => 6
 6 => 7
 7 => 8
 8 => 9
 9 => 10
)
Sum: 55
Modified Total: 55.00

Completed Scenario 2 For [ast56] [2026-02-23 00:02:02]

Render output for problem 2



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Part 2:

Progress: 100%

Details:

- Direct link to the file in the homework related branch from Github (should end in `.php` , , zero credit if it does not)
- Direct link to the file on Render.com Prod (Grab the qa url you're testing, paste it, change "-qa" to "-prod"), zero credit if , zero credit if it's not the direct link to the file or doesn't load

URL #1

https://github.com/PanicGeneral/ast56-it202-008-qa/public_html/M2/scenario2.php



URL

<https://github.com/PanicGeneral/>



URL #2

<https://ast56-it202-008.onrender.com/M2/scenario2.php>



URL

<https://ast56-it202-008.onrender.com/M2/scenario2.php>



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Part 3:

Progress: 100%

Details:

Briefly explain how your solution works. Describe the logic, steps, or decisions your code makes to reach the result. Do not restate the problem or list requirements. Additionally, did you notice anything about the math of some of the data? What do you think happened and what's the reason for it?

Your Response:

Scenario 2 was solved by just using `array_sum` to add all the values in the array and store them in a variable. Then `number_format` was used to round the numbers to 2 decimal places and I had to specify the divider to be a ".". I used `number_format` instead of `round` because `round` was outputting "1" in the case of 1.00067 instead of "1.00".



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Section #3: (2 pts.) Scenario 3 - Conversion

Progress: 100%

Task #1 (2 pts.) - Edit the `bePositive` function to make each

value positive, convert it back to the original data type, and set it to the proper slot in the `output` array

Progress: 100%

Details:

- Only make edits where noted via provided comments
- Challenge 1: Make each value positive
- Challenge 2: Convert the values back to their original data type and assign it to the proper slot of the `output` array
- Step 1: sketch out plan using comments (include uuid and date)
- Step 2: Add/commit your outline of comments (required for full credit)
- Step 3: Add code to solve the problem (add/commit as needed)

Part 1:

Progress: 100%

Details:

Two screenshots are expected

1. Snippet of relevant code showing solution (with uuid/date comment)
2. Full output of executing the program (visit the proper file on Render.com qa after a manual deploy)

```
$output = array_fill(0, count($arr), null); // Initialize output array
// Start Solution Phase
//Step 1: Iterate through the array and use absolute value to convert values

foreach($arr as $place => $val){
    $positive = abs($val);

    //Step 2: Use an if loop to determine what data type the values of the arrays are and then use type casting to change them back
    if (is_string($val)){
        $output[$place] = (string)$positive;
    }
    elseif (is_int($val)){
        $output[$place] = (int)$positive;
    }
    elseif (is_float($val)){
        $output[$place] = (float)$positive;
    }
}
```

Code snippet problem 3

```
Completed Scenario 3 for [src] [2026-02-24 00:05:14]

Scenario 3: Given a list of numbers, return a new list where each number is its absolute value.

Given:
- Input: [-1, 2, -3, 4, -5, 6, -7, 8, -9, 10]
- Expected Output: [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]

Actual:
- Output: [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]

Scenario 3: Given a list of numbers, return a new list where each number is its absolute value.

Given:
- Input: [-1, 2, -3, 4, -5, 6, -7, 8, -9, 10]
- Expected Output: [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]

Actual:
- Output: [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
```

Render output problem 3

Part 2:

Progress: 100%

Details:

- Direct link to the file in the homework related branch from Github (should end in `.php` , , zero credit if it does not)
- Direct link to the file on Render.com Prod (Grab the qa url you're testing, paste it, change "-qa" to "-prod"), zero credit if

URL #1

https://github.com/PanicGeneral/ast56-it202-008qa/public_html/M2/scenario3.php



URL

<https://github.com/PanicGeneral/>



URL #2

<https://ast56-it202-008.onrender.com/M2/scenario3.php>



URL

<https://ast56-it202-008.onrender.com/M2/scenario3.php>



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Part 3:

Progress: 100%

Details:

Briefly explain how your solution works. Describe the logic, steps, or decisions your code makes to reach the result. Do not restate the problem or list requirements.

Your Response:

I first iterated through the array and applied the absolute value to the numbers then saved them to a separate variable. That variable was then passed through an if/elseif loop to determine what data type it was and then typecasted to its correct type.



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Section #4: (2 pts.) Scenario 4 - Strings

Progress: 50%

Task #1 (2 pts.) - Edit the `transformText` function to solve the challenges

Progress: 100%

Details:

- Only make edits where noted via provided comments
- Challenges will be applied collectively

- Challenge 1: Remove non-alphanumeric characters except spaces
- Challenge 2: Convert text to Title Case
- Challenge 3: Remove leading/trailing spaces and remove duplicate spaces between words
- Result 1-3: Assign final phrase to `placeholderForModifiedPhrase`
- Step 1: sketch out plan using comments (include ucld and date)
- Step 2: Add/commit your outline of comments (required for full credit)
- Step 3: Add code to solve the problem (add/commit as needed)

Part 1:

Progress: 100%

Details:

Two screenshots are expected

1. Snippet of relevant code showing solution (with ucld/date comment)
2. Full output of executing the program (visit the proper file on Render.com qa after a manual deploy)

```
foreach ($arr as $index => $text) {
    // Start Solution Edits
    //date 02/24/2026
    //Step 1: Iterate through array and use preg_replace to remove the non-alphanumeric characters
    //Step 2: Use ucwords to make to convert strings to Title case
    //Step 3: Use trim to remove leading and trailing spaces
    $words=preg_replace("/[^a-zA-Z0-9 ]/", "", $text);
    $words=preg_replace("/\s+/", " ", $words);
    $titlecase=ucwords(strtolower($words));
    $placeholderForModifiedPhrase=trim($titlecase);

    // and solution edits
    printSummaryTransformations($index, $placeholderForModifiedPhrase, $placeholderForMiddleCharacter);
}
```

Code snippet problem 4

```

$ php -f /var/www/html/04-problem4.php
Original text: THE QUICK BROWN FOX JUMPED OVER THE LAZY DOG.
Text with non-alphanumeric characters removed: THE QUICK BROWN FOX JUMPED OVER THE LAZY DOG
Text converted to title case: The Quick Brown Fox Jumped Over The Lazy Dog
Final trimmed title case phrase: The Quick Brown Fox Jumped Over The Lazy Dog
Completed Summary for [04] [0000-00-00 00:00:00]
```

Render output problem 4

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Part 2:

Progress: 100%

Details:

- Direct link to the file in the homework related branch from Github (should end in `.php`, zero credit if it does not)

- Direct link to the file on Render.com Prod (visit the proper file on Render.com qa after a manual deploy)

URL #1

https://github.com/PanicGeneral/ast56-it202-008-qa/public_html/M2/scenario4.php



URL

<https://github.com/PanicGeneral/>



URL #2

<https://ast56-it202-008.onrender.com/scenario4.php>



URL

<https://ast56-it202-008.onrender.com/scenario4.php>



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Part 3:

Progress: 100%

Details:

Briefly explain how your solution works. Describe the logic, steps, or decisions your code makes to reach the result. Do not restate the problem or list requirements.

Your Response:

I first used preg_replace to replace every non-alphanumeric character with "". This is done by specifying that every character that is not in the pattern I provided which was all letters (upper or lower case) would be replaced by "". I then got rid of the extra spaces with the same preg_replace command but with a different pattern. The next step was to use the php function ucwords to make the strings title cases. Finally trim was used to remove the leading and trailing spaces.



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Task #2 (+ 1 pt.) - Edit the `transformText` function to solve the extra credit challenge (challenge 4, Extra Credit)

Progress: 0%

Details:

- Only make edits where noted via provided comments
- Challenge 4: Extract up to middle 3 characters (beginning starts at middle of phrase, exclude the first and last character for shorter phrases)
- Assign result to 'placeholderForMiddleCharacters'
- If not enough characters in a word, instead assign "Not enough characters" to `placeholderForMiddleCharacters`
- Step 1: sketch out plan using comments (include uuid and date)
- Step 2: Add/commit your outline of comments (required for full credit)
- Step 3: Add code to solve the problem (add/commit as needed)
- Note: There isn't a URL entry for this since it's part of Scenario 4

Part 1:

Progress: 0%

Details:

Two screenshots are expected

1. Snippet of relevant code showing solution (with ucid/date comment)
2. Full output of executing the program (visit the proper file on Render.com qa after a manual deploy)



Missing Caption

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Part 2:

Progress: 0%

Details:

Briefly explain how your solution works. Describe the logic, steps, or decisions your code makes to reach the result. Do not restate the problem or list requirements.

Your Response:

Missing Response

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Section #5: (2 pts.) Misc

Progress: 100%

Task #1 (0.67 pts.) - Github Details

Progress: 100%

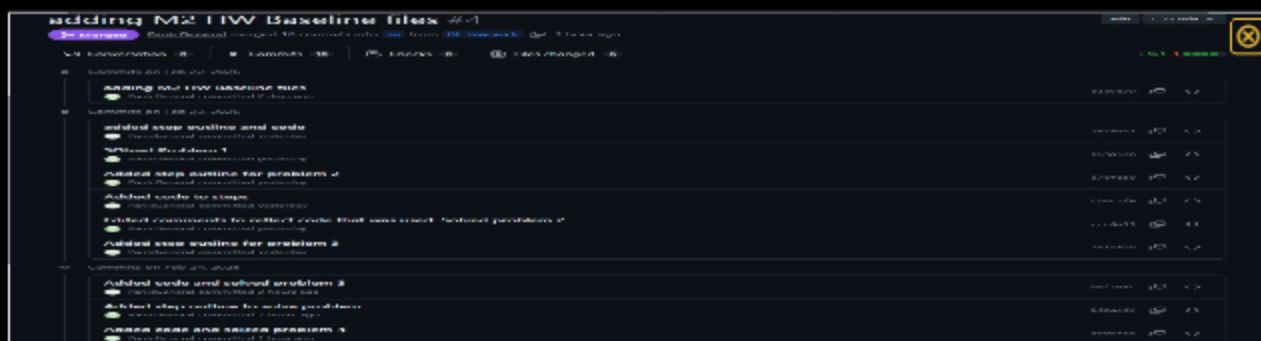
Part 1:

Progress: 100%

Details:

From the Commits tab of the Pull Request, screenshot the commit history Following minimum should be present: addition of baseline files, comments, planning thoughts, solutions

should be present: addition of baseline files, comments planning thoughts, solutions



Commit history

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Part 2:

Progress: 100%

Details:

Include the link to the Pull Request (should end in `/pull/#`, zero credit if it does not)

URL #1

<https://github.com/PanicGeneral/ast56-IT202-008/>



URL

<https://github.com/PanicGeneral/>

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Task #2 (0.67 pts.) - WakaTime - Activity

Progress: 100%

Details:

- Visit the WakaTime.com Dashboard
- Click **Projects** and find your repository
- Capture the overall time at the top that includes the repository name
- Capture the individual time at the bottom that includes the file time
- Note: The duration isn't relevant for the grade and the visual graphs aren't necessary

Projects • ast56-IT202-008

2 hrs 53 mins over the Last 7 Days in ast56-IT202-008 under all branches. 

Overall Wakatime

Files		Branches	
48 mins	public_html/scenario1.php	2 hrs 50 mins	M2-Homework
43 mins	public_html/M2/scenario4.php	3 mins	qa
29 mins	public_html/M2/scenario3.php		
27 mins	public_html/M2/scenario2.php		
16 mins	public_html/M2/scenario1.php		
3 mins	..-1-checkpoint_01-30-2026_14-03-07.pdf		
2 mins	public_html/scenario4.php		
29 secs	public_html/scenario3.php		
20 secs	public_html/base.php		
19 secs	public_html/scenario2.php		
10 secs	public_html/base		

Individual WakaTime



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☰ Task #3 (0.67 pts.) - Reflection

Progress: 100%

⇒ Task #1 (0.33 pts.) - What did you learn?

Progress: 100%

Details:

Briefly answer the question with at least a few reasonable sentences

Your Response:

I'm not a really strong programmer so I actually had to learn a lot of the php functions and how to utilize them in code. While I had a basic knowledge of them I had to research exactly how to use them to get my specific result.



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⇒ Task #2 (0.33 pts.) - What was the easiest part of the assignment?

Progress: 100%

Details:

Briefly answer the question with at least a few reasonable sentences

Your Response:

The easiest part was actually scenario 2 because it was just a simple addition and

formatting.



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⇒ Task #3 (0.33 pts.) - What was the hardest part of the assignment?

Progress: 100%

Details:

Briefly answer the question with at least a few reasonable sentences

Your Response:

I don't believe there was a specific hardest part of this assignment for me, but the most challenging was definitely having to learn the functions and figure out how exactly I can go about getting the tasks done.



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